

Two Gates, One Quantity:

How Many Populations It Takes to Subtract Sampling Variance

Abstract

Many survey methods subtract an estimate of sampling variance from the spread observed across areas or populations, to recover how much those populations truly differ. Small area estimation does this when it estimates a model variance, and so do prediction methods that calibrate on survey-estimated curves. We show that the two conditions such a procedure needs are not independent. Subtraction leaves the standard error unchanged while shrinking what it is divided by, so the design share that makes a correction worth attempting is the same quantity that makes the remainder imprecise. The algebra is classical; its consequence for deciding whether a correction may be used has not been drawn out. The requirement is a closed-form boundary: the areas needed grow as the square of the noise-to-signal ratio over the squared tolerance, so holding a Fay–Herriot shrinkage weight stable to a tenth needs two hundred areas at equal noise and signal, and 3,200 at four times the noise. We then apply the boundary to a conformal procedure that gates on exactly these two conditions and reports a fixed floor of 94 populations. That floor is the boundary at a design share of zero, where nothing needs removing; where its own need gate opens the requirement is 153. Across 45 configurations of European Social Survey estimates, spanning five ways of defining the calibration unit, the boundary predicts the gate correctly in 44 while the fixed floor does not. Refining the unit reaches the correction only along a dimension on which the domains genuinely differ: cutting by region comes within reach, cutting by sex falls an order of magnitude short.

Statement of Significance

A survey estimate is an uncertain description of a population. Using estimates from many populations to predict another introduces a second source of uncertainty, the differences between populations, and the two answer different questions. Methods exist that subtract estimated sampling variance to sharpen such a prediction. This paper shows that the conditions for using them are coupled rather than separate: the more sampling error there is to remove, the harder it is to certify what remains, because the reliability check is evaluated on the scale the correction shrinks. We give the resulting requirement in closed form and verify it on European survey data, where it predicts the method's own decision in every configuration examined while the fixed sample-size rule it replaces does not.

1 Introduction

Repeated probability surveys describe a population by a distribution rather than a mean. For an ordinal item the cumulative distribution function records the share of the population below each response threshold, and a survey programme covering many countries or many regions produces one such curve per population per round. These curves are increasingly used as inputs to a second analysis: information from the observed populations is used to construct a prediction band for a further one.

The inputs to that second analysis are estimates. Each carries sampling error from its own complex design, and a band calibrated on the discrepancies between those estimates and a prediction centre inherits that error. This paper asks what such a band guarantees, about which object, and under which assumption.

Two targets. Write $\tilde{F}_c$ for population c 's survey estimate and F_c for the population distribution behind it. A band may be asked to cover a further *survey estimate* $\tilde{F}_{K+1}$, which we call target T1, or the *population distribution* F_{K+1} , target T2. The two are not interchangeable. Exchangeability

of the observed curves delivers T1 in finite samples by the ordinary conformal rank argument. It delivers nothing about T2 on its own, because the calibration scores contain sampling error that the latent target does not.

Reaching T2 requires additional information, and there are two routes. A bound on the target's own sampling error gives a shape-free enlargement of the T1 band. Alternatively, subtracting an estimate of sampling variance from the calibration spread gives a narrower band, at the price of a distributional assumption: that the standardised observed and latent curves follow a common law. Correct coordinate variances do not imply it.

What this paper adds. We do not claim novelty for the conformal rank argument, for the observation that measurement error is not identified from a convolution, or for simultaneous scoring of a curve. The contribution is to connect these to survey-estimated distribution curves and to characterise when the variance-subtraction route is reachable in practice.

The characterisation turns on a coupling that is not new as algebra but that appears not to have been drawn out for a calibration procedure. An adaptive implementation of this correction gates on two quantities: whether there is enough design noise to be worth removing, and whether the remaining scale is determined precisely enough to trust. These read as independent requirements, and a feasibility region drawn from them looks like a corner of a plane. They are not independent. The deconvolution shrinks the target scale in proportion to the design share, and the reliability diagnostic is inflated by exactly that shrinkage. Across our survey applications the correlation between the design share and the diagnostic's inflation above its sample-size floor is 0.964.

Design noise large enough to be worth removing is, by that very magnitude, large enough to leave the remainder poorly determined. The feasible region is a narrow diagonal band, not a corner.

This displaces the account that accompanies the implementation. Its stated obstacle is a population-count floor of ninety-four, derived from the diagnostic's sampling term. In our regional applications that floor is cleared in 27 of 30 configurations, at population counts up to

290, and the correction still activates in only 4. The floor is not what binds.

Empirical characterisation. At the national unit of the European Social Survey the design share is 0.095–0.101 on full samples and reaches 0.369 in a subgroup scan; the need gate opens in none of 37 configurations. The correction is unnecessary there. One unit down, on regions within the same survey, the design share runs 0.49–0.66 and the correction becomes both needed and, in a few configurations, feasible. Where it activates it narrows the band by 15–19 percent against the uncorrected alternative, so the boundary tells a practitioner where to look rather than telling them to stop. Which configurations depends on the variance estimator: a plain m -of- m resample opens the reliability gate in 8 of 8 arms and a Rao–Wu–Yue rescaled resample in 2 of 8, with the frozen threshold falling inside that gap.

Refining the calibration unit is the natural response to a population-count requirement, and it works only in one direction. Cutting the survey into regions yields 236 populations against a requirement of 256, within reach; cutting the same survey by sex yields 60 against 2006, because men and women have nearly the same trust distribution and the split adds sampling noise without adding dispersion. Across 45 configurations spanning 5 domain definitions the boundary predicts the gate in 44 of them, and 6 open.

Finally, a simulation isolates the distributional assumption from scale estimation. With exact coordinate variances throughout, mismatched correlation structures cost simultaneous coverage in a direction that is predictable for cumulative distribution functions, and the cost grows with the number of coordinates carried simultaneously rather than shrinking with the number of calibration populations.

2 Two prediction targets

Let $c = 1, \dots, K$ index calibration populations and $j = 1, \dots, d$ a fixed stack of response thresholds, and where applicable survey rounds. Relative to a prediction centre μ ,

$$Y_c(j) = \tilde{F}_c(j) - \mu(j) = G_c(j) + S_c(j), \quad G_c(j) = F_c(j) - \mu(j), \quad (1)$$

with S_c the survey estimation error and $v_c^2(j)$ its design variance. Conditional on a finite population, design-unbiased or approximately unbiased estimation motivates a centred error; for a weighted empirical distribution the usual ratio form is not exactly design-unbiased, and we keep the idealised centred model of the theory separate from its survey approximation.

The two statements of interest are

$$\Pr\{\tilde{F}_{K+1}(j) \in \hat{B}(j) \forall j\} \geq 1 - \alpha \quad (\text{T1}), \quad \Pr\{F_{K+1}(j) \in \hat{B}(j) \forall j\} \geq 1 - \alpha \quad (\text{T2}). \quad (2)$$

Both average over the population-level model and, for observed objects, their sampling mechanisms. Neither is conditional coverage for a named population. Exchangeability is assumed of the observed curves *together with their designs and observation patterns*; a shared label such as “country” does not establish it, and unlike designs can defeat it even when the latent curves are exchangeable. Departures are a separate problem with its own framework (Barber et al., 2023).

Table 1 states what each construction may claim.

Anchor. For a centre fixed independently of the curves, let $A_c = \max_j |Y_c(j)|$, $m = \lceil (1 - \alpha)(K + 1) \rceil$ and $q_\alpha = A_{(m)}$, with $q_\alpha = \infty$ when $m > K$. Exchangeability gives T1 coverage at least $m/(K + 1)$. A radius that does not exist is reported as infinite rather than silently replaced by the sample maximum: the band is uninformative, and saying so is part of the guarantee.

Table 1: Constructions, the target each may claim, and the basis of the claim. All statements are simultaneous on a fixed finite grid.

Construction	Target	Basis
Observed anchor	T1	Exchangeable observed curves; finite-sample rank bound
Noise enlargement	T2	Anchor, widened by a bound on the target’s maximum sampling error
Oracle scale correction	T2	Common standardised-shape law (S), with known scales
Estimated correction	T2	(S), plus control of scale, centring and design-variance approximation error

Enlargement. If a deterministic $u \geq 0$ satisfies $\Pr\{\|S_{K+1}\|_\infty > u\} \leq \beta$, then a T1 band at level $1 - \alpha_0$ enlarged by u covers the latent curve with probability at least $1 - \alpha_0 - \beta$, by the triangle inequality on the intersection of the two events. No independence is needed and no shape is assumed. The route is shape-free but wide, and it is not operational for an unsurveyed population without information bounding the design error of its hypothetical survey.

Non-identification. Without information restricting S , the law of G is not identified by that of $Y = G + S$: distinct signal and noise laws generate the same convolution (Fan, 1991). This motivates the additional assumptions below. It does not establish that any particular conformal radius is optimal among latent-target procedures, and we make no width-optimality claim.

2.1 Where this sits

Conformal prediction has barely reached survey venues. *Survey Methodology* contains one conformal paper, Wieczorek (2023), which gives design-based prediction for a unit drawn from the same finite population. This journal contains two, both in one 2024 issue: Bersson and Hoff (2024), on small areas with area-specific frequentist coverage, and Michal et al. (2024), which is the closer antecedent and easy to miss because its title does not name the method. That paper proposes a scaled split-conformal procedure relaxing exchangeability for complex designs, and documents that plain split conformal undercovers there. It differs from the present setting in what

carries the sampling error: its calibration units are observations from a designed sample, whereas ours are *estimated curves*, each with a design variance of its own, and it is that error in the calibration objects rather than in the target that the correction studied here attempts to remove. The coverage-guarantee tradition this belongs to in this journal begins earlier, with Burris and Hoff (2020), whose adaptive intervals attain exact area-level coverage for every parameter value and are robust to misspecification of the linking model. The competing line, second-order-correct empirical Bayes intervals, culminating in Yoshimori and Lahiri (2014), is provably shorter than direct intervals and is the benchmark any latent-target proposal must eventually meet; we make no width claim against it, for the reason given in Section 7.

On the estimand, complex-survey estimation of a distribution function is well developed (Pasquazzi and de Capitani, 2016; Harms and Duchesne, 2006), but simultaneous bands for one are not. The only *Survey Methodology* construction of simultaneous bands for a design-based estimated curve is Cardot et al. (2013), and its curve is a functional mean rather than a distribution function. Koffman et al. (2025) is the closest existing machinery to ours—a Rao–Wu–Yue–Beaumont bootstrap supporting joint bands under a complex design—with a different estimand and no conformal component.

The resampling scheme, and a caution about it. We use the Rao–Wu–Yue rescaled bootstrap (Rao et al., 1992), which Bessonneau et al. (2021) treat for stratified multistage household surveys and Scarpa et al. (2026) report as outperforming alternatives in this journal for such designs. One objection has to be met. Saigo (2007) recommends an unrescaled, mean-adjusted scheme precisely *because* no rescaling is performed, on the grounds that rescaling sits awkwardly with estimating distribution functions and quantiles. Our estimands are distribution functions, so the objection applies. We answer it empirically rather than by assertion: Section 6 reports every diagnostic under both schemes, and the choice is consequential—the reliability gate opens in 8 of 8 arms under the unrescaled scheme and 2 of 8 under rescaling. We report the rescaled results as primary because the unrescaled scheme understates between-unit variance in a stratified design,

and we report the difference rather than absorbing it.

3 The scale correction and the assumption it needs

A narrower latent-target band is available under a stronger model. Suppose, for deterministic design scales,

$$Y_c(j) = \sqrt{s_G^2(j) + v_c^2(j)} W_c(j), \quad G_{K+1}(j) = s_G(j) W_{K+1}(j), \quad (3)$$

where $W_1, \dots, W_{K+1}$ are independent curves with a common law, zero coordinate means and unit coordinate variances. Call this assumption (S). The oracle scores $M_c = \max_j |Y_c(j)| / \sqrt{s_G^2(j) + v_c^2(j)}$ are then copies of $\|W\|_\infty$, and their conformal quantile placed on the latent scale gives a T2 band by the same rank argument.

Assumption (S) is stronger than a variance decomposition, and the difference is the subject of Section 5. Knowing $v_c^2(j)$ does not identify the cross-coordinate dependence of the latent scores. Gaussian latent curves and Gaussian sampling errors can violate (S) whenever their correlation matrices differ, because a standardised sum of independent processes need not share a summand's law. Substantively, (S) requires the sampling error to carry the same cross-threshold correlation profile as the population differences, so that noise changes only the scale of a common shape.

Write $s_Y^2(j) = s_G^2(j) + \overline{v^2}(j)$ for the observed spread and

$$\rho^2(j) = \overline{v^2}(j) / s_Y^2(j) \in [0, 1) \quad (4)$$

for the *design share*, the fraction of observed dispersion attributable to sampling. The oracle scale ratio is $s_G/s_Y = \sqrt{1 - \rho^2}$, so the correction's maximal proportional gain is $1 - \sqrt{1 - \rho^2}$, second order in ρ . That ratio is not the ratio of realised band widths, which also carries the ratio of two conformal quantiles.

Estimated scales. With estimated denominators differing from oracle values by relative error at most $\delta < 1$ outside an event of probability η ,

$$\Pr(\text{miss}) \leq \alpha + \eta + \Pr\left\{M_{K+1} \in \left(M_{(m)} \frac{1-\delta}{1+\delta}, M_{(m)}\right]\right\}, \quad (5)$$

which names the two further ingredients: control of scale-estimation error, and anti-concentration of the score law near the calibration quantile. Rates for these do not supply a numerical finite-sample level without their constants.

4 Two gates, one quantity

We take as our object an adaptive implementation of the correction that guards it with gates computed from the calibration set. Because the criterion we examine belongs to that implementation, we state it here in full rather than by reference, and everything below is reproducible from this description alone. Two gates matter. A *need* gate requires the design share to exceed a frozen cutoff, $\hat{\rho} > \rho_0$ with $\rho_0 = 0.47$. A *reliability* gate requires

$$D = \max_j \frac{\sqrt{2s_Y^4(j)/(K-1) + (\text{SD}_K(\hat{v}^2(j))/\sqrt{K})^2}}{\hat{s}_G^2(j)} \leq \tau, \quad (6)$$

where $\hat{s}_G^2$ is the guarded scale, which subtracts a one-sided margin below the mean design variance so that an uncertain estimate subtracts less. At the frozen $\tau = 0.147$ the implementation reports the requirement $K \geq 94$, obtained by noting $\hat{s}_G^2 \leq s_Y^2$, whence $D \geq \sqrt{2/(K-1)}$. That floor is correct and, as an account of what blocks the correction, misleading. The two gates are not independent requirements on two dimensions.

Proposition 1 (Gate coupling). *Let $\kappa(j) = \hat{s}_G^2(j)/s_Y^2(j) \in (0, 1]$ be the scale shrinkage and let*

$$h(j) = \frac{(\text{SD}_K(\hat{v}^2(j))/\sqrt{K})^2}{2s_Y^4(j)/(K-1) + (\text{SD}_K(\hat{v}^2(j))/\sqrt{K})^2}$$

be the share of the squared standard error contributed by dispersion of design variances across populations. Then at the coordinate attaining the maximum in (6),

$$D = \sqrt{\frac{2}{K-1}} \cdot \frac{1}{\kappa\sqrt{1-h}}, \quad (7)$$

and under (S), where the guarded subtraction recovers the design share, $\kappa \approx 1 - \rho^2$, so the reliability gate $D \leq \tau$ is

$$K \geq 1 + \frac{2}{\tau^2(1-\rho^2)^2(1-h)}. \quad (8)$$

Proof. Write $A = 2s_Y^4/(K-1)$ and $B = (\text{SD}_K(\hat{v}^2)/\sqrt{K})^2$ at the maximising coordinate, so $h = B/(A+B)$ and $A+B = A/(1-h)$. Then $D = \sqrt{A+B}/\hat{s}_G^2 = \sqrt{A/(1-h)}/(\kappa s_Y^2) = \sqrt{2/(K-1)}/(\kappa\sqrt{1-h})$, which is (7). Substituting $\kappa = 1 - \rho^2$ and solving $D \leq \tau$ for K gives (8). $\square$

The content is the placement of ρ in (8). The deconvolution shrinks the target scale by $1 - \rho^2$, and the diagnostic divides by that shrunken scale, so the same design share that opens the need gate inflates the diagnostic by $(1 - \rho^2)^{-1}$ and works to close the reliability gate. **Design noise large enough to be worth removing is, by that magnitude, large enough to leave the remainder poorly determined.**

The reported floor of 94 is (8) evaluated at $\rho \rightarrow 0$ and $h \rightarrow 0$: it is the requirement in the regime where the correction is not needed. At the need cutoff itself, $\rho = \rho_0 = 0.47$, the requirement is already $K \geq 153$; at $\rho = 0.60$ it is 227, and at $\rho = 0.66$, the largest design share in our applications, 292.

Table 2: Populations required by (8) at $\tau = 0.147$ and $h = 0$. The reported floor of 94 is the entry at a design share of zero, where nothing needs removing. The need gate opens at $\rho_0 = 0.47$.

design share ρ	0	0.30	0.47	0.60	0.66
populations required	94	113	153	227	292

Section 6 checks (7) and (8) against 30 survey configurations. The identity holds to 3.2% mean and 7.1% maximum relative error, and the boundary predicts the gate correctly in 30 of 30

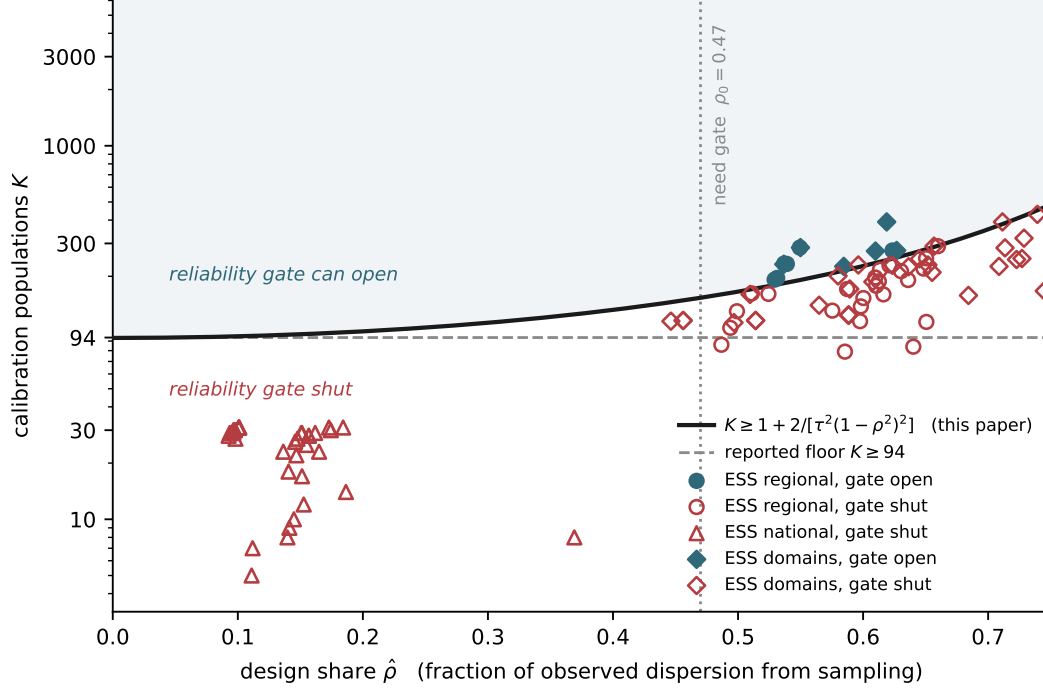


Figure 1: The feasibility boundary (8) at $\tau = 0.147$ and $h = 0$, with every survey configuration of Section 6. Filled markers are configurations where the reliability gate opened. The reported floor of 94 is the boundary's intercept at zero design share; to the right of the need cutoff the requirement rises steeply, and the region between the two rules is where every configuration that clears the floor but fails the gate lies.

Alt text: Calibration population count on a logarithmic vertical axis against design share horizontally. A rising curve separates an upper region where the reliability gate can open from a lower region where it cannot. A horizontal dashed line at 94 marks the reported fixed floor, which coincides with the curve only at zero design share and falls increasingly far below it as the design share grows. Regional survey configurations cluster between design shares of 0.49 and 0.66 at counts near 100 to 290, straddling the curve; the four with the gate open lie above it. National configurations sit far below and to the left, at counts under 35 and design shares near 0.1.

configurations, against 27 that the $K \geq 94$ rule would admit.

4.1 The inflation is an information bound

Proposition 1 is a statement about one diagnostic, built on a moment estimator. Its content is more general, and the generality matters for whether the boundary is a property of an implementation or of the estimation problem.

Subtracting sampling variance from between-population dispersion is exactly what estimating the model variance of a Fay–Herriot small-area model does (Fay and Herriot, 1979; Rao and Molina, 2015). For area c , a direct estimate $y_c = \theta_c + e_c$ with known sampling variance D_c , and

$\theta_c = x'_c \beta + u_c$ with $u_c \sim N(0, A)$. The Fisher information for A with the D_c known is

$I(A) = \frac{1}{2} \sum_c (A + D_c)^{-2}$, so any unbiased estimator obeys

$$\text{RSE}(\hat{A}) \geq \frac{1}{A} \sqrt{\frac{2}{\sum_c (A + D_c)^{-2}}} \stackrel{D_c \equiv D}{=} \sqrt{\frac{2}{K}} \cdot \frac{1}{1 - \rho^2}, \quad \rho^2 = \frac{D}{A + D}, \quad (9)$$

the same $(1 - \rho^2)^{-1}$ inflation, now from the information rather than from a particular estimator.

Requiring relative precision τ gives $K \geq 2/[\tau^2(1 - \rho^2)^2]$, which is (8) up to the unit arising from the moment estimator's $K - 1$.

Attribution. None of this is new, and it is important to say so precisely. Equation (9) is the standard asymptotic variance of the model-variance estimator in this model (Rao and Molina, 2015). The underlying phenomenon—subtracting a component leaves the standard error unchanged while shrinking what it is divided by, so relative precision degrades by the shrinkage factor—is the familiar difficulty of variance-component estimation with a small between-group component, and it is the same difficulty that makes the between-study variance imprecise in random-effects meta-analysis (DerSimonian and Laird, 1986). Partlett and Riley (2017) document the resulting coupled condition empirically in that setting: intervals behave acceptably only when the number of studies is large *and* the heterogeneity is not small relative to

within-study error, the two requirements trading off exactly as (8) describes.

What we add is not the algebra. It is three things: the transfer to a *calibration* setting, where the coupling governs whether a procedure may be used at all rather than how wide an interval is; the closed form (8), which turns the trade-off into a planning quantity evaluated at the design share actually present; and the observation, developed in Section 6, that the feasibility criterion accompanying the implementation under study is a fixed population count that is (8) evaluated where the correction is unnecessary. The implementation is stated in full in Section 4, and the authors' relationship to it is described on the title page.

Two properties of (9) matter for practice, and we verify both on 40 simulated conditions at 2000 replicates (Table 3).

The bound is attained by the estimator practitioners use. Restricted maximum likelihood tracks it closely: over conditions with $K \geq 100$ the ratio of realised to bound relative standard error has median 1.002 and range 0.952–1.035. The inflation therefore cannot be evaded by a better estimator, and the earlier restriction to unbiased estimation, which excluded exactly the shrinkage estimators small-area practice is built on, is not what carries the result.

The closed form is the equal-variance case, and it is conservative. With $D_c \equiv D$ it reproduces the exact bound to 0e+00 relative error. With sampling variances dispersed across areas at the same mean, the exact bound falls below it, by Jensen's inequality on $\sum_c (A + D_c)^{-2}$; the closed form then overstates the requirement, so using it for planning is conservative rather than optimistic.

A familiar symptom. If the boundary is real, the regime it calls infeasible should be recognisable to practitioners, and it is. The share of replicates on which REML returns the boundary estimate $\hat{A} = 0$ is 0.0% or less at design shares up to 0.5 at every K examined, and at $\rho = 0.9$ it is 25.2% at $K = 30$, 8.3% at $K = 100$ and 0.0% at $K = 500$. Zero and negative estimates of a between-area variance are a long-standing nuisance in small-area work. They arise, on this reading, in precisely the regime where the component is too small relative to sampling error for the available number of areas to determine it, which is what the boundary states.

Table 3: Relative standard error of the Fay–Herriot model variance: information bound, REML, and the share of replicates returning a zero estimate. Equal sampling variances, 2000 replicates per condition.

K	$\rho = 0.5$		$\rho = 0.9$		
	bound	REML	bound	REML	at zero
30	0.344	0.346	1.359	1.191	25.2%
100	0.189	0.183	0.744	0.708	8.3%
250	0.119	0.121	0.471	0.467	1.3%
500	0.084	0.085	0.333	0.334	0.0%

4.2 What this costs a small-area analyst

The requirement above is stated in terms of a relative-precision tolerance τ , which a gated procedure supplies explicitly. Most small-area analysis supplies none, and the natural question is what the coupling costs someone who never states one. It has a direct answer, because the estimated model variance is not the quantity of interest: it is an input to the shrinkage weight

$$\gamma = \frac{A}{A+D}, \quad \frac{\partial \gamma / \gamma}{\partial A / A} = \frac{D}{A+D} = \rho^2, \quad (10)$$

which decides how far each direct estimate is pulled toward the model. Combining the elasticity in (10) with the bound (9),

$$\text{RSE}(\hat{\gamma}) = \rho^2 \text{RSE}(\hat{A}) = \frac{\rho^2}{1 - \rho^2} \sqrt{\frac{2}{K}} = \frac{D}{A} \sqrt{\frac{2}{K}}, \quad (11)$$

the relative instability of the shrinkage weight is the noise-to-signal ratio times $\sqrt{2/K}$. Inverting it gives a planning quantity that needs no tolerance to be invented for it: to hold the weight stable to within a relative x ,

$$K \geq \frac{2}{x^2} \left(\frac{D}{A} \right)^2. \quad (12)$$

Simulation confirms (11) where the linearisation applies: over conditions with $K \geq 100$ and $\rho \leq 0.7$ the realised instability of the REML weight is 1.015 times the predicted value at the median, range 0.847–1.064. At $K = 30$ it runs about a tenth high, and at $\rho = 0.9$ the delta-method

step fails as γ approaches its boundary.

Table 4: Areas required by (12) to hold the Fay–Herriot shrinkage weight stable to within a stated relative tolerance, by noise-to-signal ratio. $D/A = 1$ is a between-area variance equal to the sampling variance of a direct estimate.

tolerance on γ	noise-to-signal D/A			
	0.5	1	2	4
20%	13	50	200	800
10%	50	200	800	3,200
5%	200	800	3,200	12,800

Table 4 is the practical content of the coupling, and it is independent of conformal prediction. It says that an area-level model whose direct estimates are as noisy as the between-area signal needs two hundred areas before its shrinkage weights are stable to a tenth, and eight hundred before they are stable to a twentieth; at four times the noise those figures are 3,200 and 12,800. The same quantity that makes shrinkage worth doing makes the amount of shrinkage hard to determine.

5 Isolating the shape assumption

Proposition 1 concerns whether the correction is *reachable*. Whether it is *sound* where reachable is a separate question, governed by (S), and coordinate variances do not answer it. We therefore hold the variances exactly correct and vary only the correlation shape.

Latent curves $G \sim N(0, R_G)$ and sampling errors $S \sim N(0, \sigma^2 R_S)$ are drawn independently with $\sigma = \rho / \sqrt{1 - \rho^2}$, so every coordinate variance is correct by construction and only R_G versus R_S is manipulated. The grid crosses coordinate counts $d \in \{8, 24, 48\}$, design shares $\rho \in \{0.29, 0.40, 0.52, 0.80\}$, calibration counts $K \in \{30, 100, 250\}$ and the six correlation pairs of Table 5, at 4,000 replicates and nominal 90%: 216 cells.

The partial-sum structure is the argument of the next paragraph made concrete rather than asserted: it is what the sampling error of an estimated distribution function actually looks like, at the threshold values this survey actually has. The first and third design shares are the maxima observed at the national and regional units in Section 6. These are standardised deviation vectors,

Table 5: The six correlation structures. All keep coordinate variances exactly correct; only the dependence across coordinates differs. “Partial sum” is $R(j, k) = \{p_j(1 - p_k)/p_k(1 - p_j)\}^{1/2}$ for $p_j \leq p_k$, the correlation of sampling error in a cumulative distribution function, evaluated at the survey’s own core threshold values $p = (0.177, 0.262, 0.370, 0.463)$ interpolated onto d coordinates. Its mean off-diagonal entry is 0.69.

condition	latent R_G	sampling R_S	role
matched	equicorr. 0.5	equicorr. 0.5	(S) holds; control
survey-realistic	AR(1) 0.3	partial sum	the mechanism, at its own values
noise more corr.	independent	equicorr. 0.8	stylised, mild
noise far more corr.	independent	equicorr. 0.95	stylised, severe
noise equi., latent AR	AR(1) 0.3	equicorr. 0.9	shape as well as strength
latent more corr.	equicorr. 0.9	AR(1) 0.3	the reverse direction

not cumulative distributions and not a finite-population design, and they supply exact scales: the experiment isolates (S) and cannot bound any procedure’s realised coverage.

The direction is predictable, and the survey-realistic direction is the damaging one. For a cumulative distribution function, sampling errors at adjacent thresholds are partial sums over one sample and are strongly positively correlated; differences between populations across thresholds carry no such mechanism. The expected configuration is therefore noise more correlated than latent, and this was recorded as a falsifiable prediction before execution. It held. Every structure with noise more correlated than latent undercovers, and the reverse overcovers, reaching 0.983 at $\rho = 0.80$, $d = 48$; a matched-correlation control sits at 0.900. The survey-realistic condition, which is not stylised at all, behaves like the mild stylised one: 0.898 at the national design share, 0.883 at the regional one, and 0.797 at $\rho = 0.80$.

The cost grows with coordinates, not with populations. At $\rho = 0.80$ and $d = 48$ the worst mismatch gives latent coverage 0.755, against 0.859 at $d = 8$. Raising the calibration count does not repair it: the same cell gives 0.776 at $K = 30$ and 0.755 at $K = 250$. The mismatch is a property of the model, not a sampling error that more populations average away.

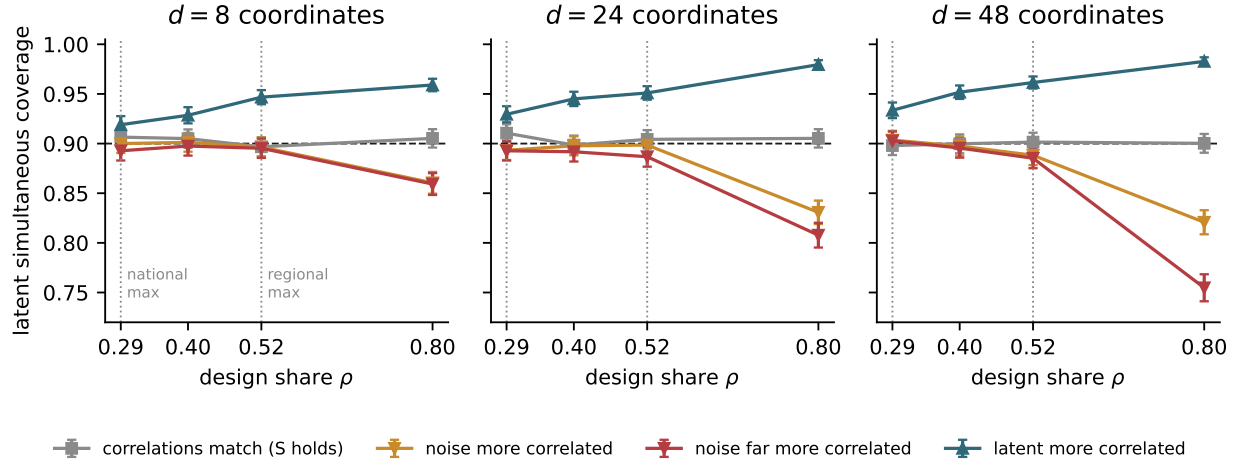


Figure 2: Latent simultaneous coverage of the oracle scale correction with exactly correct coordinate variances, by coordinate count, at $K = 250$ and nominal 90%. Bars are two Monte Carlo standard errors. Dotted verticals mark the largest design shares observed at the national and regional units.

Alt text: Three panels for eight, twenty-four and forty-eight coordinates. When the two correlation structures match, coverage stays at the nominal level throughout. When sampling error is more correlated across coordinates than population differences, coverage falls below nominal, mildly at the observed design shares and steeply at the highest one, and the fall deepens as the coordinate count rises. The reverse mismatch is conservative.

At the observed operating points the cost is small. At $\rho = 0.29$, the national maximum, no mismatch departs from nominal by more than Monte Carlo error (0.903). At $\rho = 0.52$, the regional maximum, the cost is one to two points (0.885). The severe failures require design shares above anything these surveys exhibit. The finding is therefore a warning about direction and mechanism, not a demonstration of failure in the applications reported here, and we state it as such.

6 Survey applications

6.1 Data and design

We use European Social Survey rounds 9–11, which carry primary sampling unit, stratum, region and weight variables for all 33 participating countries (European Social Survey ERIC, 2024).

Curves are weighted cumulative distributions of trust in parliament on the preregistered low-trust

core $t \in \{1, 2, 3, 4\}$, the grid of the implementation under study; since D in (6) is a maximum over coordinates, the grid is not a cosmetic choice. Design variances come from a Rao–Wu–Yue rescaled stratified bootstrap with $B = 400$ (Rao et al., 1992; Wolter, 2007).

Survey reporting. The target population is residents aged 15 and over in private households in each participating country, regardless of nationality or language. Rounds 9, 10 and 11 were fielded in 2018–19, 2020–22 and 2022–24 respectively, with country-specific fieldwork periods recorded in the provider’s documentation. The item is `trstpr1`, asked as: *“Using this card, please tell me on a score of 0–10 how much you personally trust each of the institutions I read out. 0 means you do not trust an institution at all, and 10 means you have complete trust. Firstly, ... [country]’s parliament?”* Responses of “refusal”, “don’t know” and “no answer” are excluded, as are records with non-positive analysis weight. Country-level response rates vary and are documented by the provider per country and round; we do not report a pooled rate, since no single field period or population underlies it. All estimates use the analysis weight, which combines design, post-stratification and population-size components.

The resampling unit. The regional estimand is a region’s departure from its own national distribution, and region and national curves are recomputed inside each replicate before differencing so that their covariance propagates. The resampling unit is `(stratum,psu)`, never `(stratum,psu,region)`: a primary sampling unit spanning two regions must enter a replicate whole, or the dependence the resample exists to represent is destroyed.

Whether this matters is an empirical question about the design files, and we audited all 90 country-rounds and 159,320 respondents. Sampling units nest inside regions for all but 0.52% of respondents, with violations in 15 of 65 country-rounds carrying informative units. Strata fail to nest in regions in 63 of 90. In 25 country-rounds the unit identifier is degenerate, equalling the respondent identifier: correct for individual-level register samples, but it means the pooled analysis mixes design variances that do and do not contain a clustering component, which we report separately rather than average over.

6.2 National estimates: the correction is unnecessary

At the national unit the design share is 0.095–0.101 on full samples, per round and for the stacked trajectory, and reaches 0.369 in a 33-cell subgroup scan by age band and minimum sample size.

The need gate opens in 0 of 37 configurations. Whatever else is true, there is little sampling variance to remove relative to the dispersion between countries. Consistently with Proposition 1, the dispersion term h is negligible here, 0.0–0.2%: at $K \leq 33$ the diagnostic is essentially its sample-size floor.

6.3 Regional estimates: needed, and reachable only in a narrow band

One unit down the picture reverses. Across 30 configurations, two pools by three rounds by five minimum region sizes, the design share runs 0.49–0.66 and the need gate opens in 22. The reliability gate opens in 6, and the correction activates in 4.

The sample-size floor is not the obstacle. It is cleared in 27 of 30 configurations, at counts up to 290. What binds is the coupling. The identity (7) reproduces the realised diagnostic to 3.2% mean and 7.1% maximum relative error, the correlation between the design share and the diagnostic's inflation above its floor is 0.964, and the closed-form boundary (8) predicts the gate in 30 of 30 configurations against 27 admitted by the $K \geq 94$ rule. The four activations sit where the boundary says they can: at the smallest region size, and three of four in round 11, the round with the lowest design share.

6.4 Verification

Three checks accompany the above. The conclusion does not move with the bootstrap draw (four seeds, D within 0.001) or the replicate count ($B = 200$ to 2000, D within 0.002).

The third check separates two departures from an earlier implementation of the same procedure, described on the title page. Run under that implementation's own settings, ours reproduces its reported diagnostic to four decimal places, so the comparison is between one statistic computed

two ways. The variance estimator is then decisive and the resampling unit is not: the reliability gate opens in 8 of 8 m -of- m arms and 2 of 8 Rao–Wu–Yue arms, with $\tau = 0.147$ falling inside that gap, while the resampling unit moves D by under 0.1% and never flips a gate. The latter follows from the design audit, which found 0.52% of respondents in a region-splitting unit. A downward-biased variance estimator makes a correction look reachable that a defensible one does not.

6.5 Can the boundary be outrun by refining the unit?

A practitioner facing (8) has one obvious move: refine the unit. Splitting countries into regions or demographic domains multiplies the calibration count. Refinement is not free, though. Smaller domains carry smaller samples, the design variance rises, the design share rises with it, and so does the requirement. Which side wins decides whether the boundary can be engineered around. We hold the estimand fixed as a domain’s departure from its own national distribution and vary only how domains are cut, over three rounds and three minimum sizes. Table 6 gives the result, and the answer is that the number of domains is not what matters.

Table 6: Ways to cut the same survey into domains, at a minimum domain size of 60, averaged over rounds. “Required” is (8) evaluated at the design share each cut produces.

domains cut by	K	design share	required	shortfall
sex	60	0.85	2006	−1946
age band	119	0.52	189	−70
region	236	0.60	256	−20
age band and sex	232	0.65	330	−98
region and sex	292	0.68	393	−101

The dimension matters more than the count. Splitting by sex yields 60 populations against a requirement of 2006. Splitting by region yields 236 against 256, within reach. The difference is not the number of domains, which differs only fourfold, but the design share: sex produces 0.85 and region 0.60. Halving each sample by sex adds sampling variance without adding

between-domain dispersion, because men and women have nearly the same trust distribution. Cutting by region adds both, because regions genuinely differ.

Adding a second, homogeneous dimension on top behaves the same way in miniature. Region and sex together give more populations than region alone, but the requirement rises faster still, and the cut is worse than the one it refines.

The design instruction follows, and it is the useful thing the boundary yields: **refine along a dimension on which the domains actually differ**. A cut that only subdivides a homogeneous population raises the requirement far faster than the supply, and no amount of it reaches the correction. A cut that tracks real heterogeneity can.

The boundary holds across domain definitions. Over 45 configurations spanning 5 domain definitions, three rounds and three size thresholds, (8) predicts the reliability gate correctly in 44. The gate opens in 6, 5 of them in round 11, the round with the lowest design share throughout. Those are the configurations in which the correction may be used on this survey, and Section 6.6 measures what it is worth there.

6.6 What the correction buys, and whether the anchor holds

Two questions remain, and both are answered on the same target rather than against a benchmark covering a different one.

Realised width against the anchor. The earlier implementation reported the correction's gain against a *conservative envelope*, and reported a scale ratio rather than a realised radius. Neither answers what a reader asks, which is what the correction buys over the band a sensible analyst would otherwise use. That band is the anchor, and the envelope is a weak comparator: across our configurations it is 1.63 times the anchor's radius.

Measured against the anchor at the same level on the same calibration set, the correction's realised radius is 0.805 of it at the median and 0.780–0.895 across configurations. Where the

gates actually open it is 0.805–0.849, a reduction of 15–19 percent. This sits inside the oracle ceiling $1 - \sqrt{1 - \rho^2}$, which the same configurations put at 16–22 percent, the shortfall being the ratio of the two conformal quantiles that Section 3 noted.

The correction is therefore worth having where it is reachable. That changes how Proposition 1 should be read: the boundary is not a reason to abandon the correction but the quantity that says where to look for it, and the answer at these design shares is a narrow band of configurations rather than none.

Does the anchor hold? Michal et al. (2024) document that plain split conformal undercovers under complex survey designs. Our anchor is a different construction, taking the whole curve of a population as one exchangeable unit rather than splitting units of a designed sample, but the concern transfers and is worth testing rather than arguing away.

Calibrating on all other countries’ regions and evaluating on the held-out country’s, marginal coverage of the observed target is 0.8960 at the median over 15 configurations and 0.8909–0.9061 across them, against a nominal 0.90. The largest deviation is below one standard error of a proportion at these evaluation counts, 0.029 at the smallest. The finite-sample guarantee for the observed target survives contact with a real complex design.

Coverage conditional on the held-out country does not, and is not claimed: the worst country in each configuration runs 0.10–0.60. This is the ordinary marginal-versus-conditional distinction of conformal prediction, and it is the reason the guarantee is stated over populations rather than for a named one. A reader who needs a guarantee for one country in particular should use an area-level construction designed for it (Burris and Hoff, 2020; Bersson and Hoff, 2024), not this band.

6.7 What the framework delivers: one band, a family of claims

The results so far are about scope: when a correction may be used, and when it may not. A scope result needs a companion showing what the framework yields when applied, and the available one is a survey-inference finding rather than a substantive one.

For a country with observed rounds, take the contrast surface $\theta_{a,b}(t) = F_b(t) - F_a(t)$ over ordered round pairs and core thresholds. Rounds are independent cross-sections, so each round’s sampling units are resampled within its own strata. One studentised one-sided sup- t critical value covers the whole surface, so a family of claims of increasing strength—some adjacent pair declines across the core, the span declines, every adjacent pair declines—is read off one coverage event, with no multiplicity spent per claim. Certified claims hold jointly with probability at least $1 - \alpha$, by set inclusion in the band, and the count of claims does not enter.

Table 7 applies this to trust in parliament over European Social Survey rounds 9–11 at $\alpha = 0.10$, in 30 countries.

Table 7: Countries certified, by claim and by how uncertainty is handled. The first column is the sign of the point estimate and carries no inference; the second adds design-based uncertainty pointwise; the third adds simultaneity over the whole contrast surface. Trust in parliament, rounds 9–11, $\alpha = 0.10$.

claim	point estimate	pointwise	simultaneous
some adjacent pair declines	20	15	11
the span declines	10	7	6
every adjacent pair declines	4	2	1

Two reductions compound, and they are worth separating because they answer different questions. Of the 20 countries whose point estimates fall in some adjacent pair, 15 survive design-based uncertainty applied one contrast at a time, and 11 survive a band covering the whole surface at once. Roughly a quarter of the apparent movement is within sampling error, and a further fifth of the original is within the multiplicity of looking at every pair and threshold. The claim rung then compounds again: 11 countries certify some adjacent decline, 6 certify the span (AT, BE, EE, GB, GR, NL), and 1 certifies decline at every adjacent pair.

We record that the earlier implementation reported the first and third columns of the first row as a contrast between a “marginal reading” and a simultaneous band. The first column is not a marginal statistical reading; it is the sign of an estimate with no uncertainty attached. The decomposition above is the honest form of the same comparison, and it remains a large effect.

What this does not establish. Nothing here is a claim about political change. The window contains a pandemic round and a mode transition in several countries, either of which produces the shifts these bands certify, and measurement comparability across rounds is assumed rather than shown. The statement is about response distributions under those assumptions. Its purpose here is to size what simultaneity and design-based uncertainty cost a repeated cross-national reading, which is a survey-methodological quantity.

7 Discussion

Prediction from survey-estimated curves requires naming the target. Exchangeability of observed curves yields a finite-sample guarantee for a further survey estimate, and nothing more. Reaching the population distribution requires either a sampling-error bound, which is shape-free and wide, or a scale correction, which is efficient and rests on a distributional assumption that coordinate variances cannot verify.

Where the correction is reachable is governed by Proposition 1, and the answer is narrower than a population-count threshold suggests. The quantity that makes a correction worth attempting is the quantity that makes its result hard to certify, because the diagnostic is evaluated on the scale the correction itself shrinks. A feasibility region drawn as “enough populations, and enough noise” is a diagonal band, and the reported floor of 94 is its intercept at a design share of zero, where nothing needs removing.

The survey evidence is consistent throughout. National estimates sit in the unnecessary regime. Regional estimates within the same survey generate real sampling noise, and mostly fail the reliability requirement at the very design shares that make them candidates. Whether the few surviving configurations activate depends on the variance estimator, which is not a reassuring place for a feasibility result to rest.

What we do not claim. No universal width-optimality for any latent-target band; the non-identification argument motivates assumptions rather than certifying a radius. No

demonstrated improvement over existing noisy-calibration or small-area methods, since no matched comparison was run. Activation of a selector is not evidence that (S) holds: the gates read scale precision, and a procedure that abstains has not thereby proved its approximation accurate anywhere else. Coverage measured against survey estimates does not validate coverage for latent curves, and where the latent object is unobserved, as in the regional application, that check is unavailable in principle rather than merely absent.

For practice. Three levels should be reported separately and never interchanged: a finite-sample level for the observed target, a latent-target level with the assumptions and error terms that support it, and an empirical coverage assessment under a stated simulation design. Alongside them belong the unit assumed exchangeable, the grid, and the branch a selector returned. Where a correction is contemplated, (8) gives the population count required at the design share actually present, which is a more useful planning quantity than a fixed floor.

Open problems. Informative latent-target guarantees under weaker shape restrictions than (S), since Section 5 shows the cost of violating it is directional and grows with simultaneity. A version of (8) for structured covariance across populations, which the equal-variance closed form does not address and which the dispersed-variance results of Section 4.1 show is not a small effect. And design-variance estimation that does not force a choice between a downward-biased scheme under which a correction appears reachable and a defensible one under which it does not.

Data and Code Availability

Code, protocols, a claims ledger binding every reported number to the artefact that produced it, and the outputs of every experiment are in the replication repository. The simulation of Section 5 requires no microdata. European Social Survey microdata are distributed under the provider's access conditions and are not redistributed.

Acknowledgements and Analysis Planning

Each experiment has a protocol written and committed before execution, stating the outcomes that would count against the hypothesis. The sensitivity analysis of Section 5 was designed after methodological development and is exploratory; its full grid is reported. One earlier version of the regional analysis reported a result that a verification run overturned, and the protocol now requires a reimplementation to reproduce an inherited statistic before any disagreement with it is reported.

References

- Barber, R. F., E. J. Candès, A. Ramdas, and R. J. Tibshirani (2023). Conformal prediction beyond exchangeability. *The Annals of Statistics* 51(2), 816–845.
- Bersson, E. and P. D. Hoff (2024). Optimal conformal prediction for small areas. *Journal of Survey Statistics and Methodology* 12(5), 1464–1488.
- Bessonneau, P., G. Brilhaut, G. Chauvet, and J. Garcia (2021). With-replacement bootstrap variance estimation for household surveys: principles, examples and implementation. *Survey Methodology* 47(2), 313–347.
- Burris, K. C. and P. D. Hoff (2020). Exact adaptive confidence intervals for small areas. *Journal of Survey Statistics and Methodology* 8(2), 206–230.
- Cardot, H., A. Dessertaine, C. Goga, É. Josserand, and P. Lardin (2013). Comparison of different sample designs and construction of confidence bands to estimate the mean of functional data. *Survey Methodology* 39(2), 283–301.
- DerSimonian, R. and N. Laird (1986). Meta-analysis in clinical trials. *Controlled Clinical Trials* 7(3), 177–188.
- European Social Survey ERIC (2024). ESS rounds 1–11, integrated files (Data Wizard subset). <https://ess.sikt.no>. Subset documented in the replication package

(docs/DATA_SOURCES.md). Integrated-file edition DOIs: 10.21338/ess1e06_7, 10.21338/ess2e03_6, 10.21338/ess3e03_7, 10.21338/ess4e04_6, 10.21338/ess5e03_6, 10.21338/ess6e02_7, 10.21338/ess7e02_3, 10.21338/ess8e02_3, 10.21338/ess9e03_3, 10.21338/ess10e03_3, 10.21338/ess10sce03_2 (round-10 self-completion file), 10.21338/ess11e04_2; sample design data files 10.21338/ess7sddfe1_2 and 10.21338/ess8sddfe01_1.

Fan, J. (1991). On the optimal rates of convergence for nonparametric deconvolution problems. *The Annals of Statistics* 19(3), 1257–1272.

Fay, R. E. and R. A. Herriot (1979). Estimates of income for small places: an application of james-stein procedures to census data. *Journal of the American Statistical Association* 74(366), 269–277.

Harms, T. and P. Duchesne (2006). On calibration estimation for quantiles. *Survey Methodology* 32(1), 37–52.

Koffman, L. et al. (2025). Function-on-scalar regression with complex survey designs. arXiv:2511.05487.

Michal, V., J. Wakefield, A. M. Schmidt, A. Cavanaugh, B. E. Robinson, and J. Baumgartner (2024). Model-based prediction for small domains using covariates: a comparison of four methods. *Journal of Survey Statistics and Methodology* 12(5), 1489–1514.

Partlett, C. and R. D. Riley (2017). Random effects meta-analysis: coverage performance of 95% confidence and prediction intervals following reml estimation. *Statistics in Medicine* 36(2), 301–317.

Pasquazzi, L. and L. de Capitani (2016). A comparison between nonparametric estimators for finite population distribution functions. *Survey Methodology* 42(1), 87–120.

Rao, J. N. K. and I. Molina (2015). *Small Area Estimation* (2nd ed.). Wiley.

- Rao, J. N. K., C. F. J. Wu, and K. Yue (1992). Some recent work on resampling methods for complex surveys. *Survey Methodology* 18(2), 209–217.
- Saigo, H. (2007). Mean-adjusted bootstrap for two-phase sampling. *Survey Methodology* 33(1), 61–68.
- Scarpa, S., M. R. Ferrante, and S. Sperlich (2026). Inference for the quantile ratio inequality index in the context of survey data. *Journal of Survey Statistics and Methodology* 14(3), 844–864.
- Wieczorek, J. (2023). Design-based conformal prediction. *Survey Methodology* 49(2). Statistics Canada, Catalogue No. 12-001-X. arXiv:2303.01422.
- Wolter, K. M. (2007). *Introduction to Variance Estimation* (2nd ed.). New York: Springer.
- Yoshimori, M. and P. Lahiri (2014). A second-order efficient empirical bayes confidence interval. *The Annals of Statistics* 42(4), 1233–1261.